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A systematic review of Kansei engineering in vehicle design

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ABSTRACT

This study focused on the application of Kansei engineering (KE) in the field of vehicle design. Quantitative and qualitative analyses of relevant studies in the literature were conducted using knowledge graphs and bibliometric analysis tools to aid researchers and other relevant readers better understand the application of KE in the field of vehicle design. In this study, 562 research papers from the Web of Science core collection were selected, and several visual knowledge maps were generated using CiteSpace and VOSviewer for clustering and other statistical analyses. Based on these maps, this study systematically summarized the development and evolution of KE applications in the fields of vehicle design, research trends, research hotspots, key research groups, and their interrelationships, which provide important reference values for future work. Finally, three scientific issues surrounding KE were proposed: design feature deconstruction of the product, quantitative characterization of Kansei, and construction of the Kansei decision system. The core technology of the KE was discussed, the methods and challenges of KE in scientific research and practical applications were clarified, and a theoretical basis and technical support for future design and development was formulated. Thus, this study provides a systematic reference and guidance for future research and applications of KE in the field of vehicle design.

1. Introduction

The Japanese term Kansei is the subjective perception of products by the user, which is represented by comprehensive feedback, such as sensations, impressions, and emotions. It is a type of tacit knowledge that maps the user's requirements for the product [1]. Kansei Engineering (KE) places the user at the center, can create a dynamic and orderly balance between rigorous science and technology and divergent creative design. Further, it solves the problem of fuzzy and uncertain design information in the conceptual design stage. It creates a unified design language for different types of product research and development to facilitate design communication, evaluation, and decision-making. Consequently, it has been widely adopted in industrial design, interaction design, service design, and other fields.

KE originated in the field of transportation and was proposed by professor Mitsuo Nagamachi [2]. Transportation product design includes various products such as vehicles (cars, planes, ships), transportation infrastructure (stations, terminals, roads, bridges), accessories, and related services (public transportation systems, ride-sharing platforms) [3]. KE has been applied in all these fields, among which research on vehicle design has been the most extensive [4]. The links between the vehicles and people are extremely close. The design of vehicles should consider the comfort and safety of users while focusing on their Kansei needs. KE integrates human sensibility into the design process through scientific methods and can create vehicles that are in line with people's emotional and psychological expectations.

However, the research roadmap for the application of KE in the field of vehicle design remains unclear. Moreover, there is a lack of statistical analysis and systematic research guidelines for research directions [5]. Thus, this study used Knowledge Graph (KG) methods and bibliometric analysis tools to conduct both quantitative and qualitative analyses. Consequently, visual KGs were generated to help researchers and other interested readers better understand the application of KE in the field of vehicle design [6]. To conduct a comprehensive analysis, 562 research papers were selected from the Web of Science core collection. The selected studies were analyzed using clustering and other statistical functions using two well-known bibliometric software tools, CiteSpace and VOSviewer [7]. By integrating the analysis results, a set of synthetic visual KGs was obtained [8]. Based on the comprehensive analysis, the research hotspots and trends of the KE in the field of vehicle design were

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identified and discussed. This study aimed to address the following questions:

(1) What are the developments, evolutions, and research trends in the application of KE to the field of vehicle design?
(2) What are the research hotspots, key research groups, and their inter-relationships with KE applications to the field of vehicle design?
(3) What are the core technologies in KE scientific research and practical applications to the field of vehicle design?
(4) What are the KE future research directions to the field of vehicle design?

The remained of this paper is organized as follows. In Section 2, we present the methodology used in this study for the bibliography analysis. Section 3 presents an application to the field of vehicle design. The primary results and key technologies are discussed in Section 4.

2. Methodology

Our research focused on KE in vehicle design. We collected data from January 1992 to December 2023 and used techniques, such as keyword mapping, clustering, and statistical analysis, to identify research trends and hotspots.

2.1. Research objectives

Objective 1: To determine the number of publications in the Web of Science core collection focused on the application of KE in vehicle design between January 1992 and December 2023.
Objective 2: To conduct data analysis using Citespace and VOSviewer to obtain visualization graphs.
Objective 3: To analyze graphs to obtain research trends, hotspots, and future directions for the application of KE in vehicle design.
Objective 4: To explore the core technology of the KE, systematically clarify the methods and challenges of KE in scientific research and practical applications, and provide a theoretical basis and technical support for its future design and development.

2.2. Analytical methods and tools

A KG is a structured data model representing a digital framework that embodies knowledge through concepts and their connections [9]. In a KG-based data model, entities are nodes or vertices, whereas the connections between them are edges or links. This type of representation is widely used across various disciplines. To conduct a multidimensional scientometric analysis of the literature, scholars are developing various types of visualization software such as Bibexcel, CiteSpace, IN-SPIRE, VOSviewer, and CoPalRed. CiteSpace contains powerful literature statistical analysis functions and relatively large adjustable parameters that

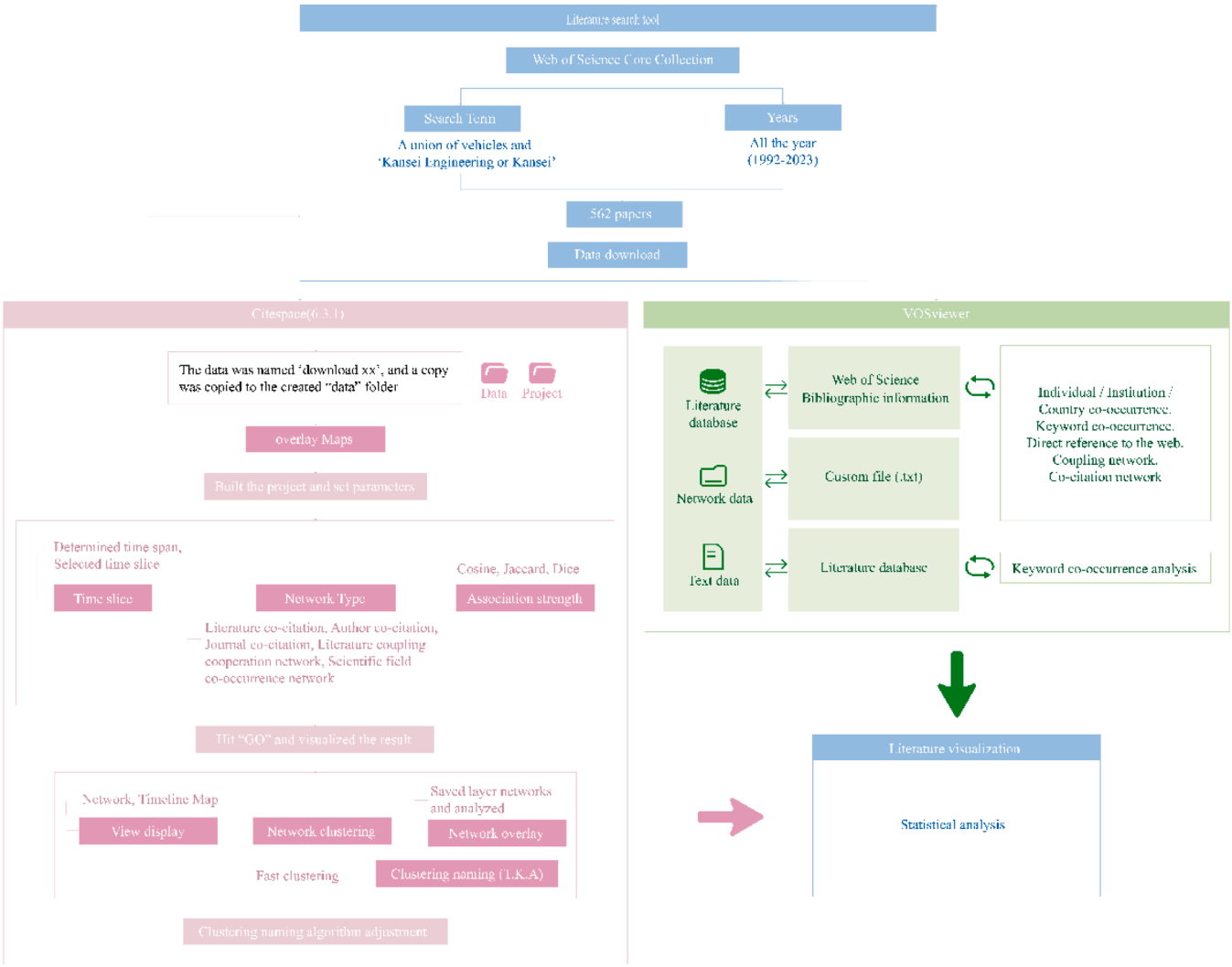


Fig. 1. Methodological framework.

can generate scientific and beautiful literature visualizations [10]. VOSviewer has superior node- clustering capabilities and can quickly generate cluster graphs [11]. To create and integrate meaningful visual KGs, this study used CiteSpace and VOSviewer to support the statistical analysis. The combination of the two tools facilitated a more comprehensive literature visualization with more dimensional data and various forms of visual diagrams, which contributed to a better understanding of the results of the statistical analysis.

Fig. 1 presents the general framework of the literature analysis. The data in this study were gathered from the Web of Science core collection and papers were collected from January 1992 to December 2023. The combination of ``Kansei Engineering'' or ``Kansei'' and various transportation terms, such as satellite, aircraft, aerospace, car, metro/subway, bicycle/bike, ambulance, boat, ship, and submarine, were used as search keywords. After deduplication, a database containing 562 articles was obtained.

The database, comprising 562 papers, was imported into CiteSpace and VOSviewer for analysis. In CiteSpace, imported data were renamed according to certain rules to obtain overlay maps and build projects. We then set the relevant parameters in CiteSpace, which were the selection of time slices, the selection of network type (including literature co-citation, author co-citation, journal co-citation, etc.), and the selection of association strength. Subsequently, various types of visual analysis results can be obtained such as networks, timelines, and clustering maps, which can be adjusted to achieve the best display effect. In VOSviewer, different data forms can be selected to obtain different types of visualization, such as keyword co-occurrence analysis, individual/institution/country co-occurrence analysis, and co-citation analysis. Finally, the results were combined to conduct a comprehensive statistical analysis to generate a simplified and multi-dimensional visual KG that can display and summarize research trends and frontier hotspots.

3. The application of KE in vehicle design

3.1. Development stage analysis

The time zone is a view of knowledge evolution from the time dimension, which clearly shows the updating and interaction of literature. In this type of map, the size of a node represents the frequency of keyword occurrence. When nodes are displayed as rings, the widths of the rings represent the number of keywords that appear in a particular year. The position of the node represents the year wherein the keyword first appeared. The line between the nodes indicates the existence of a reference link between two keywords. The links between keywords can reflect evolutionary paths, thereby helping scholars discover new keywords that emerge over time. If the time-zone function of CiteSpace is used to analyze the keywords of KE in the field of vehicle design, a time-zone map of the evolution path is obtained, as shown in Fig. 2.

As shown in Fig. 2, the content and frequency of keywords exhibited three stages of change. Before 2005, the nodes of keywords possessed the characteristics of the largest radius and obvious purple outer rings, which was the origin stage of KE. After 2018, the keyword nodes showed a small yet numerous trend, which can be divided into new stages of divergence and mergence. The intermediate period is the transitional phase of development.

1992–2005: KE was in its early stages, and related concepts were proposed in the automotive industry. KE originated in professor Mitsuo Nagamachi's study of emotion technology, which was published in Ergonomics in 1974 [12]. In 1986, Kenichi Yamamoto, President of the Mazda Motor Corporation, changed the term emotion technology to KE when he presented a keynote speech on automotive culture at the University of Michigan. In this phase, the research considers cars as the primary object [13] and constructs a mapping model based on the idea of engineering to investigate the relationship between design features, user behavior, and user feedback. Scholars have also focused on the physical and psychological aspects of users and user preferences to assist designers in decision making.

2006–2017: The application of the KE was further extended to transportation systems and plays a role in other types of vehicles such as airplanes [14] and trains [15]. In this phase, algorithms were introduced to help preprocess the user data and create models to analyze the data. Neural networks [16], fuzzy logic [17] and support vector regression [18] are the most commonly used algorithms. Using the developed hardware tools, researchers collected physiological data, such as blood

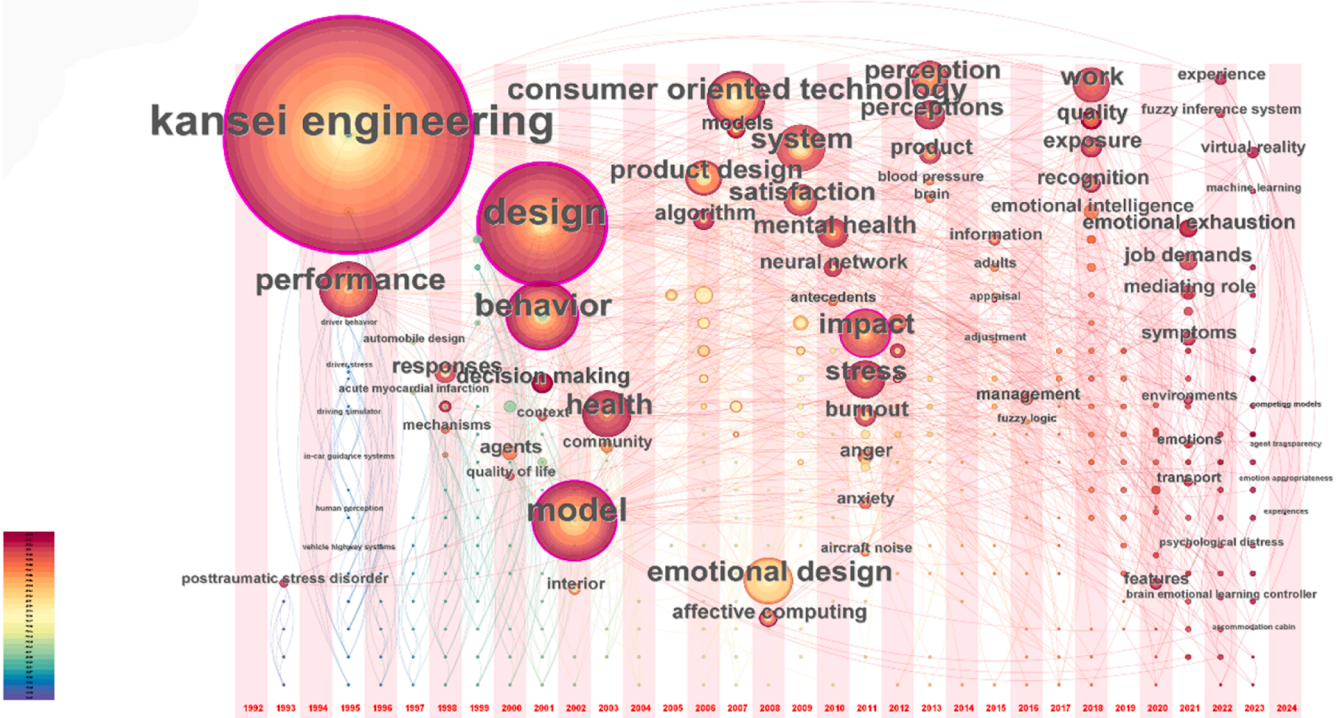


Fig. 2. Timezone map of keywords.

pressure and electroencephalograms (EEG) [19,20], combined with the detection and analysis results of common emotions, such as stress, anger, and fear. Further, they relied on algorithmic tools to create models to quantify Kansei. Scholars began to focus on the psychological well-being of users and proposed concepts of emotional design and emotional computing. In general, current research focuses on the psychological and physiological stresses experienced by users when they interact with different modes of transportation. The explosive application of algorithms overcomes the limitations of the early statistical use of regression. This breakthrough has increased the number of data sources and further improved computing power.

2018 to present: At this stage, KE research has shown a trend of simultaneous divergence and merging, and its application in transportation is expanding to other means, such as ships [21]. KE has been applied to a broader range of subdomains, such as environmental interaction, cognitive fatigue, emotional intelligence, and privacy protection. Further, KE is being integrated with computer science, biology, and other disciplines to form a new research model. At this stage, the application of the algorithms is gradually maturing. For example, fuzzy logic promotes the formation of a complete fuzzy reasoning system [22]. Simultaneously, KE continues to introduce new technologies, such as virtual reality [23] and machine learning [24], and to develop technologies surrounding emotional intelligence, such as brain emotional learning controllers [25]. Researchers continue to focus on emotional appropriateness and improving the quality of user work when using different means of transportation by relieving psychological pressure and reducing emotional fatigue to provide reliable methods and data for the optimal design of transportation products.

3.2. Core journals analysis

Core journals in the field of research are considered to be highly professional, informative, and of high quality. They can present the state of development in the field and attract the attention of scholars [26].

Double-graph overlay analysis was performed using CiteSpace to visualize the distribution of published journals on the map of journals, as well as their evolution and integration. The citation curve is shown in Fig. 3. The left side of the map shows the citing journal, the right side shows the cited journal, the two ovals represent the specific position of the citing journal in the journal map, the horizontal axis represents the number of authors, and the vertical axis represents the number of published papers [27]. The cited journals represent the application field of existing research, whereas the cited journals represent the technical support of existing research. The co-occurrence of the two can show the theoretical basis and development trend of the existing research on the application of the KE in the field of vehicle design. Curves ① and ② point from computer science and psychology toward mathematical science, respectively. Thus, existing research is focused on the study of mathematical algorithms and quantitative research on user psychology, and is supported by the development of computer science. Curve ③ points from physiological discipline to psychological discipline, indicating the continuous development of the discipline itself and promoting the application of KE in the field of vehicle design. This figure shows the disciplinary evolution of the application of the KE in vehicle design.

Generally, core journals have high citation frequencies [28]. Analysis of the literature data indicated that the three journals and proceedings with the top three publications were the International Journal of Industrial Ergonomics, AHFE 2021, and Journal of Business & Industrial Marketing, and the top three citations were the International Journal of Industrial Ergonomics, Journal of Business Research, and Computers & Industrial Engineering. When analyzing the journals, the existing research papers were divided into three main categories: safety and health, integration with computer science, and industrial engineering and commerce.

3.2.1. Safety and health

Safety and health are important topics in vehicle design research. The application of the KE can quantify users' Kansei and identify users' perceptual needs to better adapt transportation means to users' behaviors and cognitive laws. This effectively improves safety, and protects their physical and psychological health.

3.2.2. Integration with computer science

Computer Science provides technical support for intelligent transportation through technologies that support machine-learning algorithms, such as the Internet of Things, Computer Vision, and Natural Language Processing [29]. Advances in Computer Vision, Sensor Technology, and Image Processing have facilitated autonomous vehicles in accurately perceiving the surrounding environment, including the detection of roads, obstacles, and other vehicles and improving the understanding and responsiveness of autonomous driving systems to the surrounding environment [30]. Machine Learning algorithms

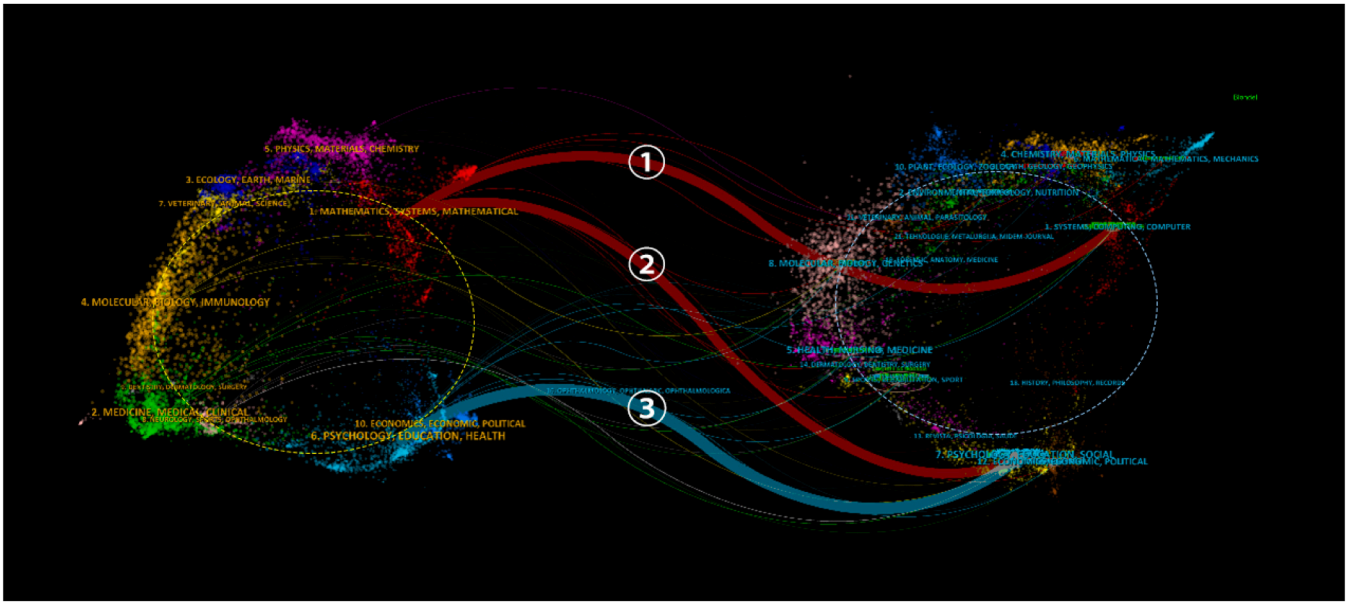


Fig. 3. Dual-map overlay.

continuously optimize the performance of autonomous driving systems through big data analysis and learning. This results in better driving decisions and behaviors [31]. Computer Science provides key technologies and methods for software engineering, system design, and verification, which can help improve the safety and reliability of autonomous driving systems and reduce accident frequency.

With the integration and development of computer science and psychology, KE has become increasingly intelligent and systematic; for example, using tools to collect user physiological data and combining algorithms to create models to predict user perceptions. The development of computer science has impacted further progress in KE.

3.2.3. Industrial engineering and commerce

As a complex industrial product, the design and production process of transportation involves many industrial problems, such as ergonomics and engineering design. Cars and other means of commodity transportation are also involved in business strategies, marketing, and other business areas.

Core journals can help identify research hotspots and suitable research topics. The relevant research is highly interdisciplinary, spanning design, computer science, industrial engineering, and business.

3.3. Core countries analysis

The betweenness centrality is an index that defines the importance of nodes in a network. CiteSpace uses this index to determine the importance of the literature (circled in purple). This index measures the importance of a node in a network, based on the number of shortest paths through the node. If the number of times a node acts as an intermediary is high, its betweenness centrality will also be high. This index is used in CiteSpace to measure the importance of literature and highlight the literature (authors, journals, institutions, etc.) with a purple circle. As shown Fig. 4, the seven countries: China, the United States, the United Kingdom, Germany, France, Australia, and Sweden, have distinct outer rings. This implies that their betweenness centrality is strong and that they have strong transnational research links and serve as indispensable bridges for multilateral collaborative research.

The number of publications and centrality between countries/regions are shown in Fig. 5, which lists the 10 most important countries/regions in terms of publication volume. Among them, China, the United States, the United Kingdom, Germany, and Australia reappeared, indicating that their publication volumes and betweenness centralities were at the top. Their research on KE in the field of vehicle design has produced relatively rich results and has a strong joint effect, and the degree of cooperation and communication is close. Compared to other countries, China and the United States have a larger node radius, more

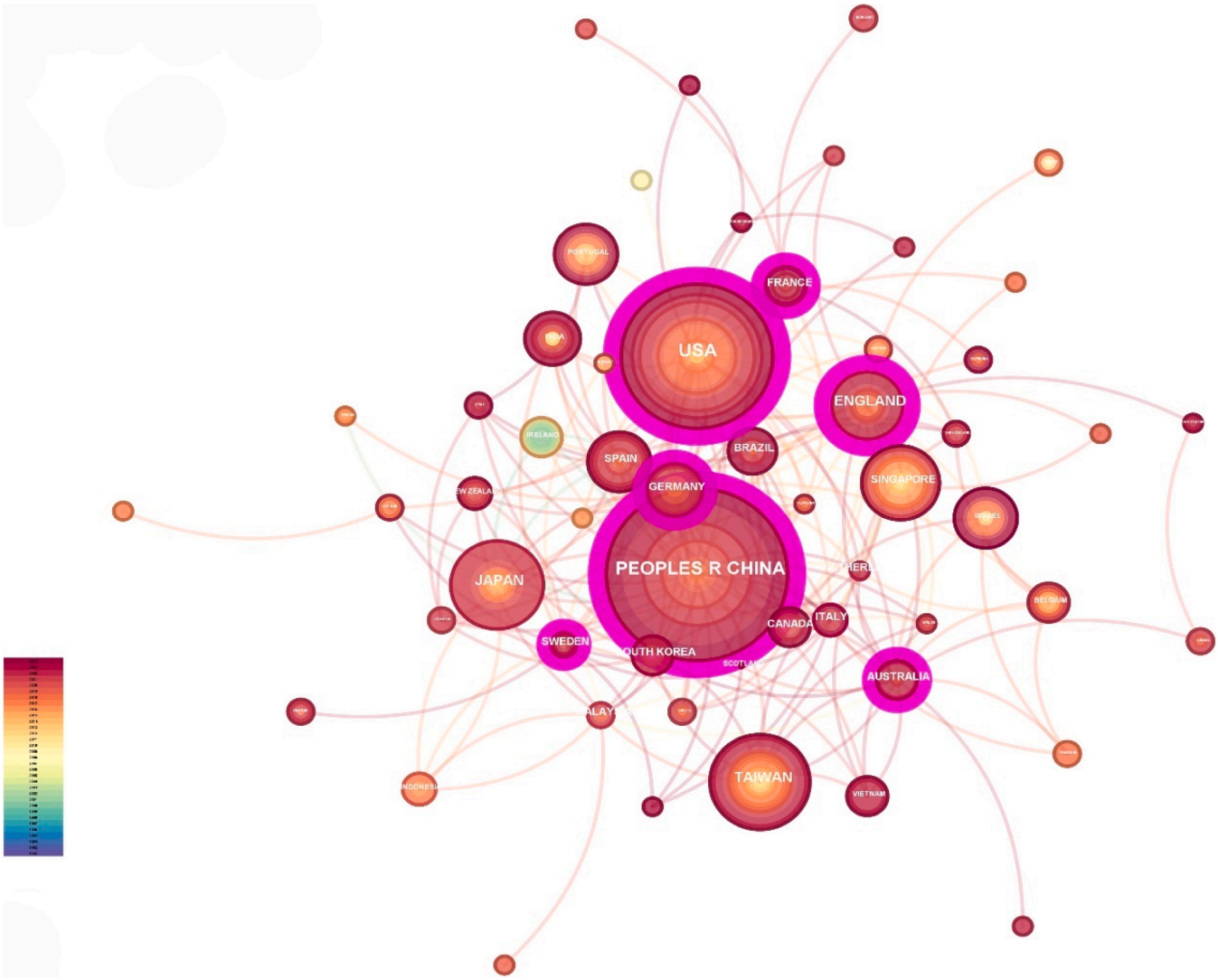


Fig. 4. National cooperation map.

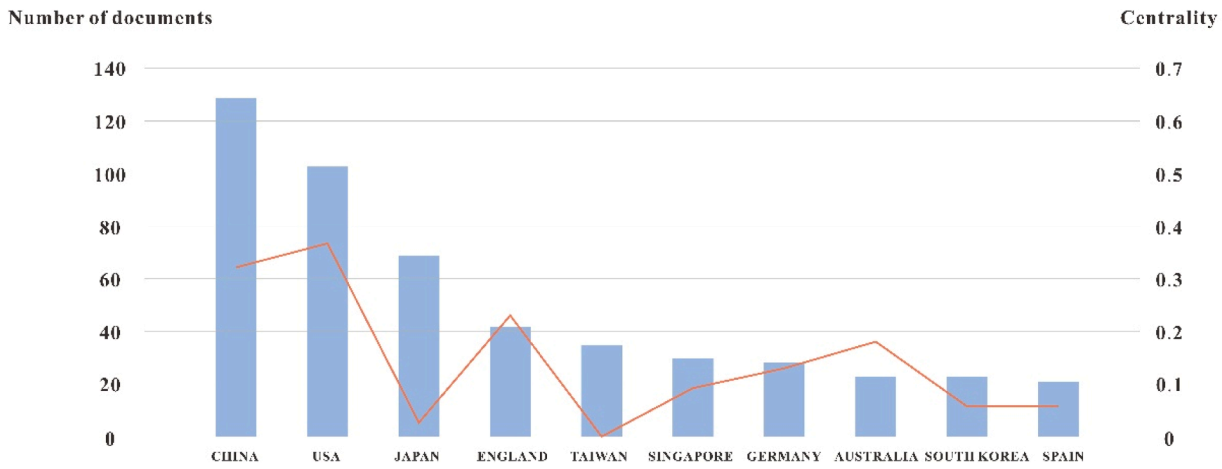


Fig. 5. The number of documents and centrality of core countries.

papers, greater influence, and are at the international leading level. In addition, Japan, the country with the third-largest number of publications, remains an important research country in this field, as it is the country where the concept of KE was born and the country that first introduced KE to the field of transportation. However, Japan's betweenness centrality was only 0.03, which is extremely low among the top 10 countries. Thus, search in Japan is relatively independent, with less cross-national cooperation and collaboration with other countries/regions.

3.4. Core authors analysis

Knowing the core authors and their citations in a research field is conducive to mastering the subject basis, research hotspots, and frontiers in this field.

Based on the statistical analysis of the number of academic papers published by researchers in the fields of KE and vehicle design, Table 1 lists the top 10 researchers in this field and their research areas. The analysis showed that researchers in this field have not yet formed an obvious cooperative network. Their research fields mainly focused on computer science and biology in two broad categories. Research can be roughly divided into three directions: land, sea, and air.

In the field of land vehicle design, Chen, Chun-Hsien [32] and Chang [33] separately studied bicycles. Both were committed to improving the user experience by optimizing the KE methodology and applying it to the saddle design for bicycles and e-bikes. Aoyama combined KE with robust design methods to optimize the exterior design of a car [34]. Rui studied the optimal design of buses [35].

In the field of aviation, Jafari et al. studied the optimal design of drones by introducing the concept of brain–emotional learning [36].

In the field of water vehicle design, McCartan et al., optimized the design of boats, including yachts [37].

Thus, as evident, the introduction of new concepts such as brain emotional learning and the cross-integration of methods in multiple fields can promote the innovation and optimization of KE's own methods, thereby providing new tools and guidance for the optimization design of various products in the field of vehicle design.

Statistical analysis of the literature cited in academic papers published in the fields of Kansei and vehicles is presented in Table 2. Table 2 lists the top 10 cited researchers in this field and their mediating centrality.

As shown in Table 2, Nagamachi, the researcher who first proposed the concept of KE and applied the KE method to the field of vehicle design, had a significantly higher number of citations and intermediary centrality than the other nine researchers. Thus, the research framework and methods proposed by him have had a profound impact on subsequent research and have laid the foundation for the development of KE. The remaining nine researchers were broadly divided into three categories. The first group of researchers focused on supplementing and applying KE methods. For example, Ishihara [38] and Hsiao [39] focused on applying KE methods to design various products, including automobiles and robots. Guo [40], Jindo, T [41] and Shieh [42] introduced tools such as genetic algorithms, BP neural networks, and rough set theory into KE, and tested them through product design examples. The second type focused on the exploration of consumer psychology from a market perspective. One of the research directions, as focused on by Hair [43] and Fornell [44], was business and economy. They focused on exploring the factors influencing consumer satisfaction. The third group of researchers focused on the regularity of human behavior and emotions. Russell JA [45] focused on children as research subjects to explore the relationship between basic human emotions and facial expressions. Podsakoff [46] focused on the study of behavioral perceptions and work stress of people.

Thus, KE, as an interdisciplinary discipline, is extremely inclusive, and its application fields and subject basis are extensive; therefore, it has strong academic vitality.

Table 1
Author publication number.

Ranking	Author	Count	Research direction
1	Chen,Chun-Hsien	7	bicycle and electric bicycle
2	Mohammad Jafari	4	UAV
3	Hideki Aoyama	4	car
4	Xu, Hao	4	UAV
5	Chao, Fei	3	brain emotional learning control
6	Danni Chang	3	bicycle
7	McCartan, Sean	3	boat
8	Carreira, Rui	3	bus
9	Duan, Qijun	3	boat
10	Deana McDonagh	3	boat

Table 2
Author citation number.

Ranking	Author	Count	Centrality
1	Nagamachi M	86	0.16
2	Ishihara S	20	0.01
3	Hsiao SW	18	0.02
4	Guo F	17	0.00
5	Hair JF	17	0.05
6	Jindo T	16	0.02
7	Shieh MD	15	0.01
8	Fornell C	14	0.04
9	Russell JA	13	0.03
10	Podsakoff PM	13	0.01

3.5. Analysis of KE application in land, air and sea

3.5.1. KE and land vehicle

In the early phase of development, KE's primary research object was land vehicle in the form of automobiles. With progress in science, technology, and economic development, cars have gradually become mass products. Compared to other means of vehicle, cars are used more frequently and are closely associated with individual users. Consumer experiences and feelings have gradually become dominant factors in car design. By focusing on users' emotional responses and preferences, KE helps develop products that appeal to consumers at a psychological level and optimize the overall user experience. Using KE, automotive products fulfil functional needs and evoke certain emotional responses that improve customer satisfaction and loyalty. This ultimately increases competitiveness in the market. For example, Mazda used KE to design exquisite and beautiful cars, which gave consumers the impression of low fuel consumption, and was very popular in the US market during the outbreak of the oil crisis.

For the application of KE in the field of land vehicle design, existing research has primarily focused on three areas: interior, appearance, and human-computer interaction. Research on interior design has mainly investigated evaluation methods of interior design. Nagamachi et al. used KE to develop a comfort diagnosis system based on design elements and dimensions to evaluate the appearance of vehicle interiors, thus pioneering the application of KE in the field of vehicle interiors [47]. Song et al. designed a vehicle interior CMF (Color, Material and Finishing) based on KE and eye-tracking technology and conducted a more scientific evaluation of the interior design through human visual and tactile perception [48].

Researchers have emphasized the relationship between user sensitivity and product characteristics. For example, Nagamachi et al. conducted a KE study on the design of car fronts in Japan, the United Kingdom, and the United States. They found similarities and differences between different countries in terms of perceived meaning and structure [49]. Lai et al. combined KE to propose a hybrid apriori + structural equation model (SEM) system to quantify design features and perceptual ratings and further identify users' perceptual requirements for the appearance of new-energy vehicles [50]. Kang combined KE and rough set theory to investigate the relationship between customer visual sensitivity and the design of a Hybrid Electric Vehicle (HEV) form factor to assist designers [51].

With respect to human-computer interaction, the design of information systems in vehicles has become the heart of human-computer interfaces. Starting from touchscreen interactions to voice or gesture recognition, the influence of different types of interactions is an important area of research. Yang et al. clarified the influence of representative modes on the driving experience and their significance levels based on an analysis of factor loadings and multiple regression equations [52]. Park et al. employed KE to evaluate the effectiveness of the design of vehicle early warning systems and additional information under low-visibility conditions. They focused on the design, evaluation, and updating of specific products and mechanisms [53]. Chen et al. proposed a design process that combined KE, rough set theory, and back-propagation neural networks to design the human-machine interface of self-driving cars for the elderly to satisfy the emotional needs of elderly users [54]. Lai et al. proposed jointly analysis of users' online comments and offline in-cabin usage behaviours, and developed the NLP-MDCEV (natural language process—multiple discrete-continuous extreme value) model to efficiently fuse these two datasets for mining users' Kansei requirements for the intelligent connected vehicle functions [55].

The application of KE in land transportation vehicles is pioneering and divergent. This is primarily reflected in the diversification, intelligence, and systematization of methods and the richness of research objects, thereby reflecting the inclusiveness and vitality of the KE. At present, consumers pay more attention to the emotional value of a product and the psychological experience of using it. This highlights the importance of the KE in the automotive sector.

3.5.2. KE and air vehicle

Since 2006, KE has been introduced in the field of airplanes and has been widely used and validated. Existing studies have mainly considered manned airplanes and other types of airplanes as research subjects.

Manned aircraft environments are complex and dynamic. They face high risks, serious accident consequences, and high design, production, and maintenance costs. Therefore, their design is increasingly focused on safety. Data show that more than 70 % of aviation accidents are caused by human factors [56,57], of which pilot accidents are the most important [58]. The situational awareness and decision-making skills of pilots are directly related to flight safety. In addition to professional training, the logic of interaction in the cockpit and atmosphere in the cabin are of crucial importance and facilitate the prevention and reduction of aviation accidents [59]. In recent years, increasing attention has been paid to the influence of psychological and emotional factors on the performance of pilots. Researchers worldwide have examined the design of aircraft cockpits from the perspective of perceptual design, mainly considering three aspects. ① Specific design application for cockpit modeling. For example, Liu traced the images of cockpit modeling and simplified them according to the importance of the lines to extract the average shape, which provides references for the creativity of cockpit modeling [60]. ② Building a mapping model between local cockpit design features and perceptual evaluation to achieve the purpose of prediction. For example, Wang et al. used the image of the T-shaped area of the cockpit as the visual stimulus, the semantic differential method to obtain the perceptual evaluation of the subjects, the body analysis method to deconstruct the design elements of the cockpit, and the BP neural network to build a correlation model between the design elements and the perceptual evaluation to predict the value of the perceptual evaluation of the new samples [61]. ③ By applying data intelligence and machine learning, a computer-aided design system for the shape or color scheme of the cockpit design was developed to provide decision support for the perceptual evaluation of the cockpit interior. Further, based on KE, Chen et al. developed a comprehensive intelligent color evaluation system for aircraft cockpits using fuzzy set theory and a support vector machine. Thus, by training and testing the evaluation model, the perceptual evaluation of the cockpit color scheme in the cockpit can be realized [62].

The cockpit is the area operated by the pilot and the cabin is the area where passengers can rest. The aircraft cabin is an enclosed and confined space with high altitude, closed, overweight, and weightless air circulation, and other special features. Untrained passengers can easily feel uncomfortable in such environments for a long time; therefore, improving their comfort and satisfaction is an important research topic. Yang et al. combined KE and virtual reality and proposed a design method for cabin interior and in-flight equipment considering the aspects of safety, passenger demand, CMF, and airlines to effectively improve the comfort of users [63]. A comfortable flight environment helps reduce passenger stress and anxiety during the journey and improves the airline's reputation and competitiveness.

Studies on other aircrafts have mainly focused on their appearance. For example, Li et al. proposed a data-driven intelligent service model, bio-inspired design (BID), based on KE, which considered the users' perceived needs and applied it practice when designing the appearance of UAV products [64]. Huang et al. applied KE to the multilayer support platform of a rocket to obtain the best color design scheme [65].

Compared to land vehicles, the complex and dynamic environment of airplanes necessitates attention on the psychology of users. Here, the application of KE helps designers understand the impact of design features on the psychology of pilots and passengers. The result is a design that better satisfies emotional needs and improves comfort and safety.

3.5.3. KE and water vehicle

Compared with other means of transportation, such as cars and

airplanes, research into the use of KE in watercraft began much later. It is only in recent years that research on KE in ship cabins has been conducted. Research has addressed the lack of emotional effect in the design of accommodation cabins and the development of a design system to reduce operator fatigue and stress [21].

The environment wherein ships operate differs from a general terrestrial environment. The marine environment is characterized by high salt content and humidity; therefore, ship-related products must be extremely resistant to corrosion, moisture, and mold [66]. In addition, a ship is cut off from the outside world when it sails at sea for long periods, and the hull sways, tilts, and shakes. These restrictions imply that people on board must work and live in an environment with limited resources, limited areas of activity, and the risk of storms. Against this background, it is particularly important to focus the psychology and emotions of a ship's crew, researchers, and other personnel, which is also an important future research direction for the KE in the field of watercraft [67].

3.6. Research hotspot and development trend analysis

3.6.1. Research hotspot analysis

The keywords: High conciseness of the author's corresponding research, can fully reflect the core of the literature. Keyword co-occurrence analysis was conducted to examine keywords with high frequency and high centrality in the literature and CiteSpace and VOSviewer were used to visualize them, as shown in Fig. 6. Fig. 6a and b show the two forms of the keyword co-occurrence graphs. In Fig. 6a, the size of the nodes represents the frequency of keyword occurrence and the connection between nodes represents the co-occurrence relationship of keywords. Further, the purple outer circle represents the high mediation centrality of the keyword, indicating the importance of the keyword in the study, such as design, KE, model, influence, performance, and behavior. Fig. 6b shows a visualization of the density of keyword co-occurrences. Areas with high density are shown in yellow, representing the most frequent keywords such as product, user, system, group, and support.

Keyword clustering is a network cluster formed using keywords related to similar research topics in the research field. CiteSpace provides two indexes, modularity(Q) and silhouette(S), according to the network structure and clarity of clustering. They can be used to judge the effect of map drawing. In general, a Q value greater than 0.3 implies that the divided structure is significant. The clustering results are considered convincing when the S-value reached 0.7. As shown in Fig. 7, the parameters of the clustering effect in this study were $S = 0.8841$ and $Q = 0.6933$, indicating a significant cluster structure and high reliability. After clustering, the primary research hotspots of KE in transportation Were #0 Kansei Engineering, #1 Emotional Health, #2 Product Design, #3 Behavior, #4 Mentation, #5 Public Transportation, #6 Boat Design, #7 Unmanned Aircraft, #8 Accidents, and #9 Therapy. These ten hotspots can be further categorized into three aspects: product design, user Kansei, and user behavior.

Product design: The concept of KE was first applied to automotive design. Related research mainly focuses on the interior, appearance, and human-computer interaction, effectively improving the level of car styling and design experience. Later, its application gradually expanded and was combined with other fields of transportation product design, such as ship and aircraft design. Owing to its versatility, KE has developed different focuses and methods when combined with different fields. In the field of ship design, because of the special environment of ships, KE is more focused on the psychological and emotional health of ship users and considers on the emotional design of ships and related products. In the field of aircraft design, KE has different focus areas of UAV, aircraft cabins, and aircraft cockpits. KE UAV research has focused on identifying user preferences to support shape optimization. Further, KE research on aircraft cabins has focused on the influence of color, lighting, and other factors on user comfort. When studying cockpits, researchers have focused on optimizing the shape and color of the cockpit to reduce the psychological pressure on drivers.

User Kansei: This research focuses on users' Kansei changes when interacting with transportation products. It employs relevant KE methods to detect and analyze users' psychological states, providing new ideas for product design. In the early and middle stages, KE mainly studied users' emotional feedback on the design features of complex products, such as cars, to promote sales, reduce stress, and reduce fatigue. With the emergence of new concepts such as emotional intelligence and technologies such as machine learning and virtual reality, KE research has gradually expanded. The internal mechanism of user emotion generation and fluctuation was explored to predict user feedback more accurately and reduce the time and cost of conducting user experiments.

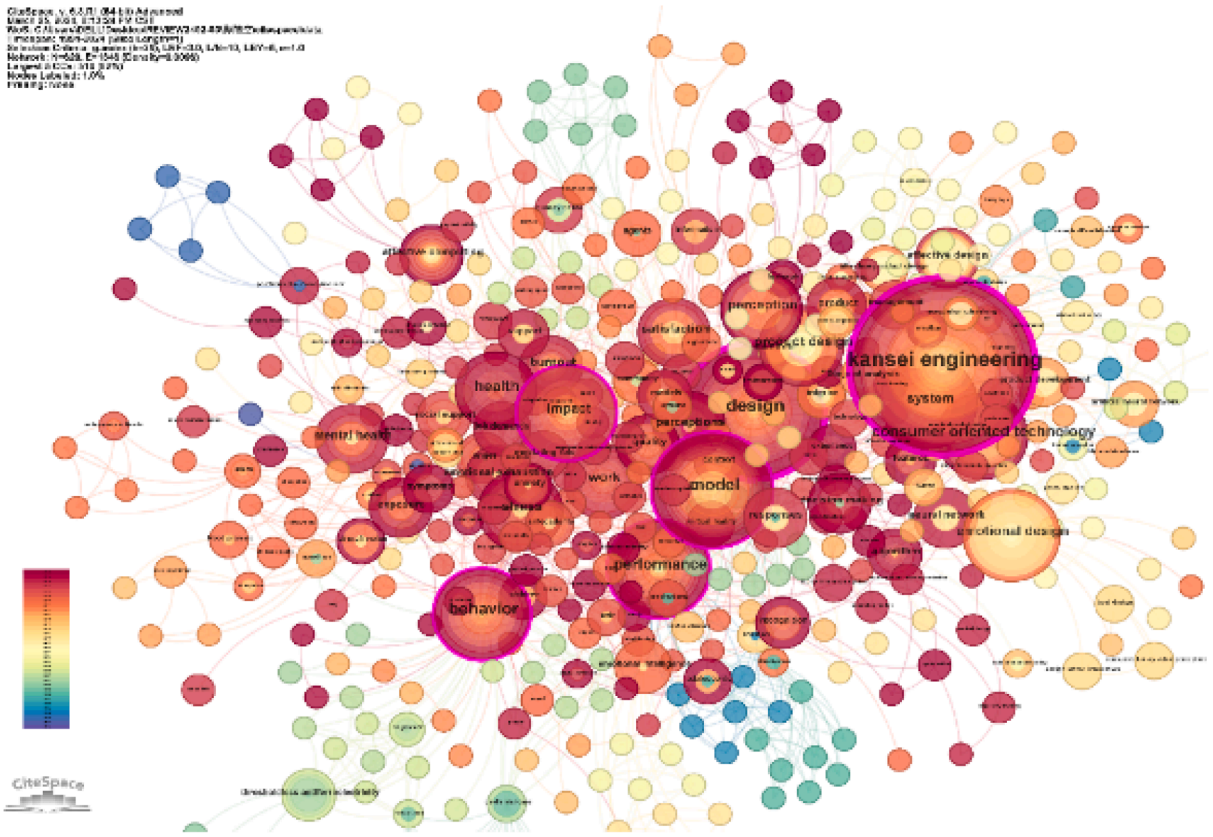
User behavior: User interviews, questionnaires, user experience maps, personas, storyboards, and other KE methods can be used to explore potential pain points and feasible solutions when using products. In transportation, vehicles are typically considerably more complex and difficult to use than other products. Moreover, there are more touchpoints when interacting with users. As transportation products serve as space carriers for accommodating and transporting users, the consequences of errors or improper operations are extremely serious. Therefore, KE has focused on user behavior, optimizes product design according to ease of use, operability, and other requirements to reduce the likelihood of accidents. In addition, the health and recovery of users after injury have also been considered.

3.6.2. Analysis of research trends

The timeline visualization map of the keyword clustering of cited documents can be used to predict the development trends of disciplines and interdisciplinary integration. In Fig. 8, the right column shows the keyword cluster labels, and the keywords under the label are arranged from right to left according to the time order. The larger the circle of a keyword, the more frequently documents containing it were cited, and these were important or landmark documents in the field. The links between the nodes represent the citation relationships between documents. The number and strength of the links reflect the degree of mutual influence and association between documents. Additional links indicate that a document is closely related to other documents. KE has been active since it was first proposed. It has been widely used in the field of vehicle design and remains an important research topic. In 1987, the first reference appeared in KE clustering, marking the starting point of this line of research. Since 2005, there has been a significant increase in the research results of this clustering, reaching a peak after 2010. This indicates that increasing attention has been paid to the application of KE in the field of vehicle design. Throughout the development process, 2005 and 2010 were the key years for the emergence of iconic literature, which had a significant impact on the development of fields with high citation or high mediation centrality. Further, the topics that have been active in recent years are conjoint analysis, fear, emotional exhaustion, emotional needs, blood pressure, public health and brain emotional learning controller. These can be further divided into three categories: methods and tools, user Kansei, and safety and health.

The development and rise of big data, artificial intelligence, brain-computer interfaces, and other technologies and their combination with KE have facilitated a trend towards intelligence and systematization of methods and tools in the field of KE. With the continuous emergence of new technologies, the methods and tools in the field of KE will continue to iterate, innovate, and develop. The acquisition, quantification, and application of user Kansei continue to be important research topics when using the KE for transportation. With the innovation of methods and tools, issues such as more accurate and efficient extraction of user Kansei, and more scientific and rigorous quantification of user Kansei, will continue to lead to its development as a research hotspot. Safety and health are unavoidable topics of continuous attention in the field of vehicle design. In-depth research on user Kansei can facilitate better adaptation of products to users' cognitive laws,

(a)



(b)

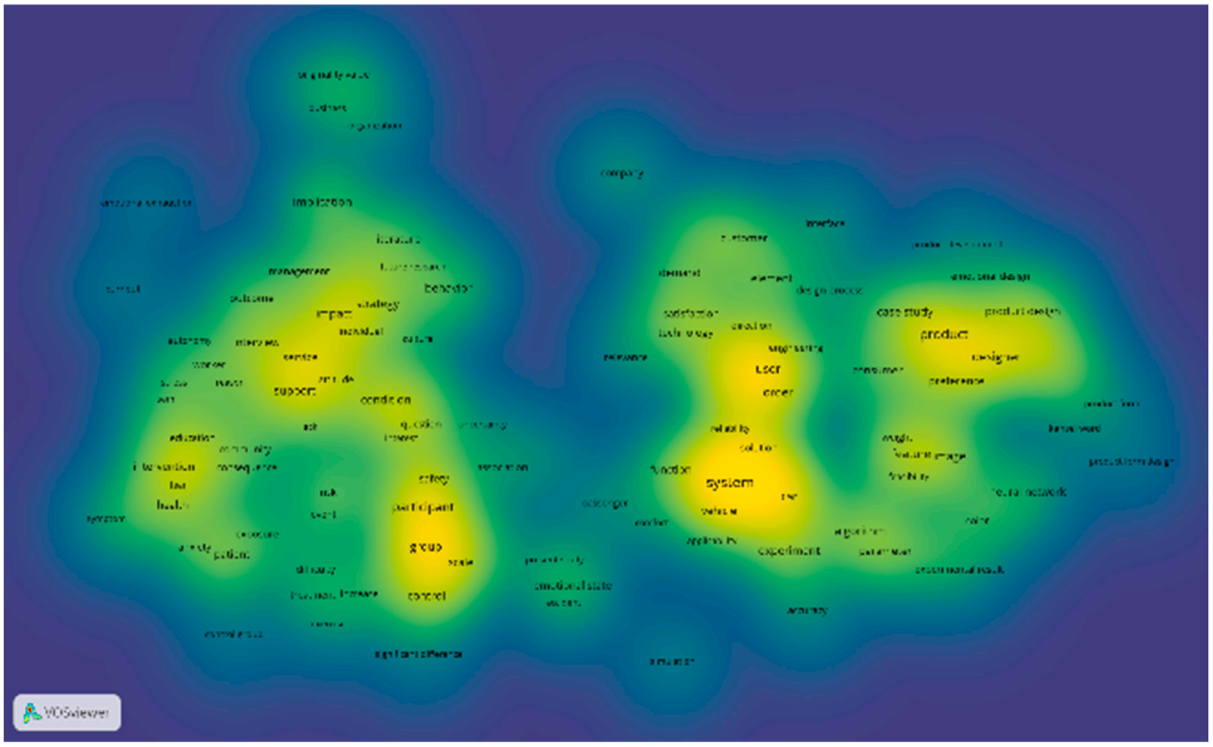


Fig. 6. (a). Keywords co-currence. (b). Keywords co-currence.

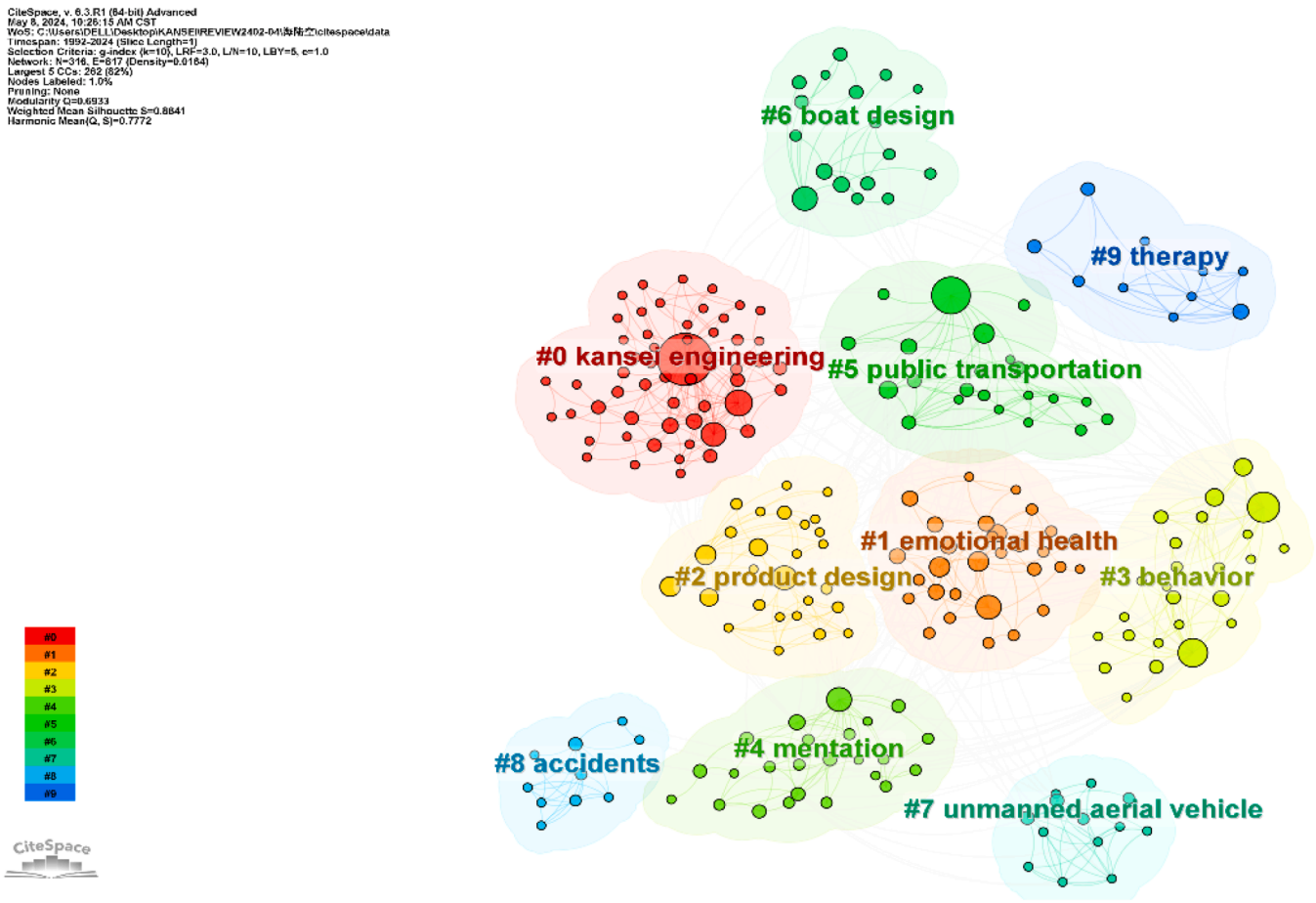


Fig. 7. Keywords clustering.

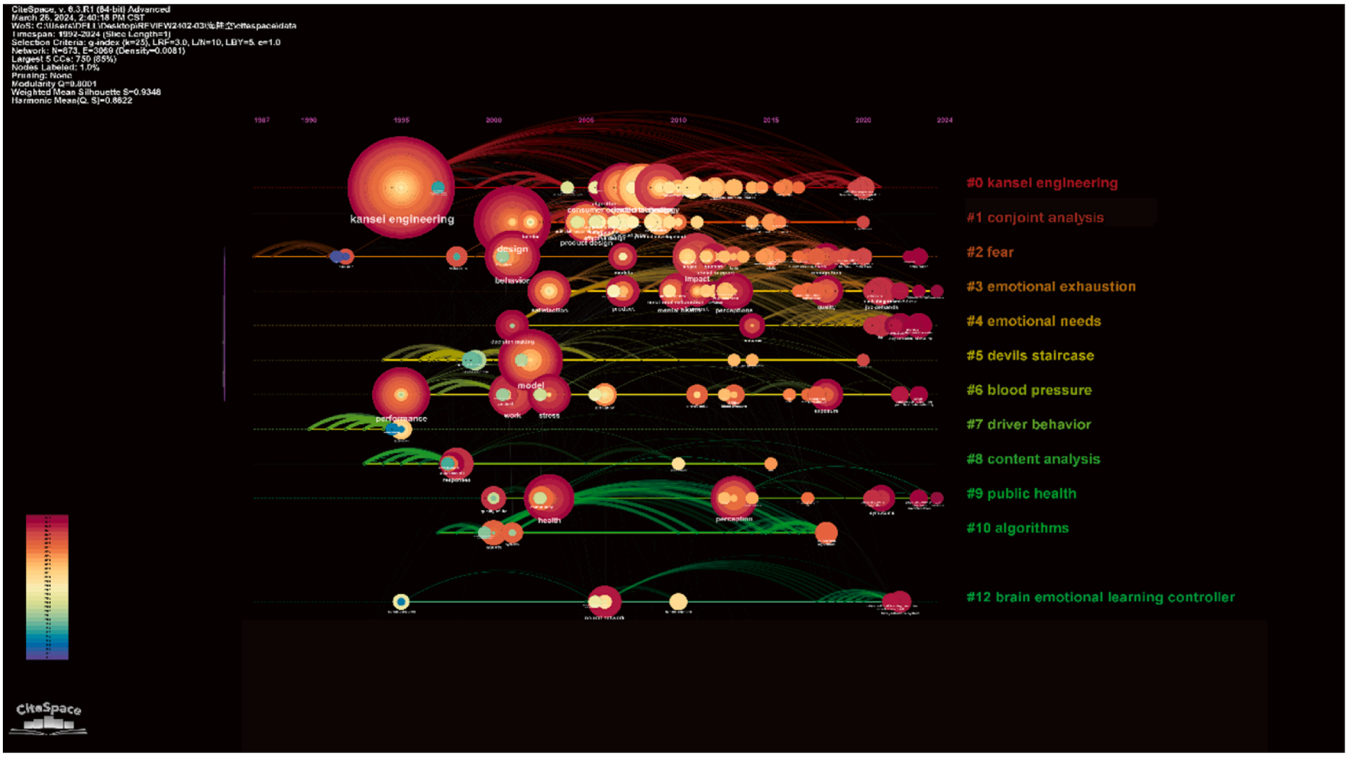


Fig. 8. Timeline visualization map of keywords clustering.

improving their comfort and safety, and thus playing a positive role in users' psychological and physical health.

4. Discussion about key technologies

KE focuses on three actors: the product, user, and designer. It comprises three scientific problems: the deconstruction of the product's design features, quantifying the representation of the user's Kansei, and the construction of a system decision model, as shown in Fig. 9. Deconstructing design characteristics involves fully understanding and capturing products, whereas the user's Kansei represents the design requirements. By constructing the system decision model considering both aspects, reliable, visible, and reusable design knowledge can be obtained, and the cognitive gap between the design team and user can be closed.

4.1. Deconstruction of product design features

The appearance of a product serves as a medium of communication between the user and designer and manifests itself in a structured collection of design features that determine the design mechanism of the product. User requirements are implemented based on product features. The shape, proportion, curvature, color, and texture of a product can influence a user's Kansei. Understanding the characteristics of product design is particularly important in KE, and forms the foundation for subsequent research.

KE follows the cognitive topology of the whole and deconstructs the design features of products based on the following cognitive principles.

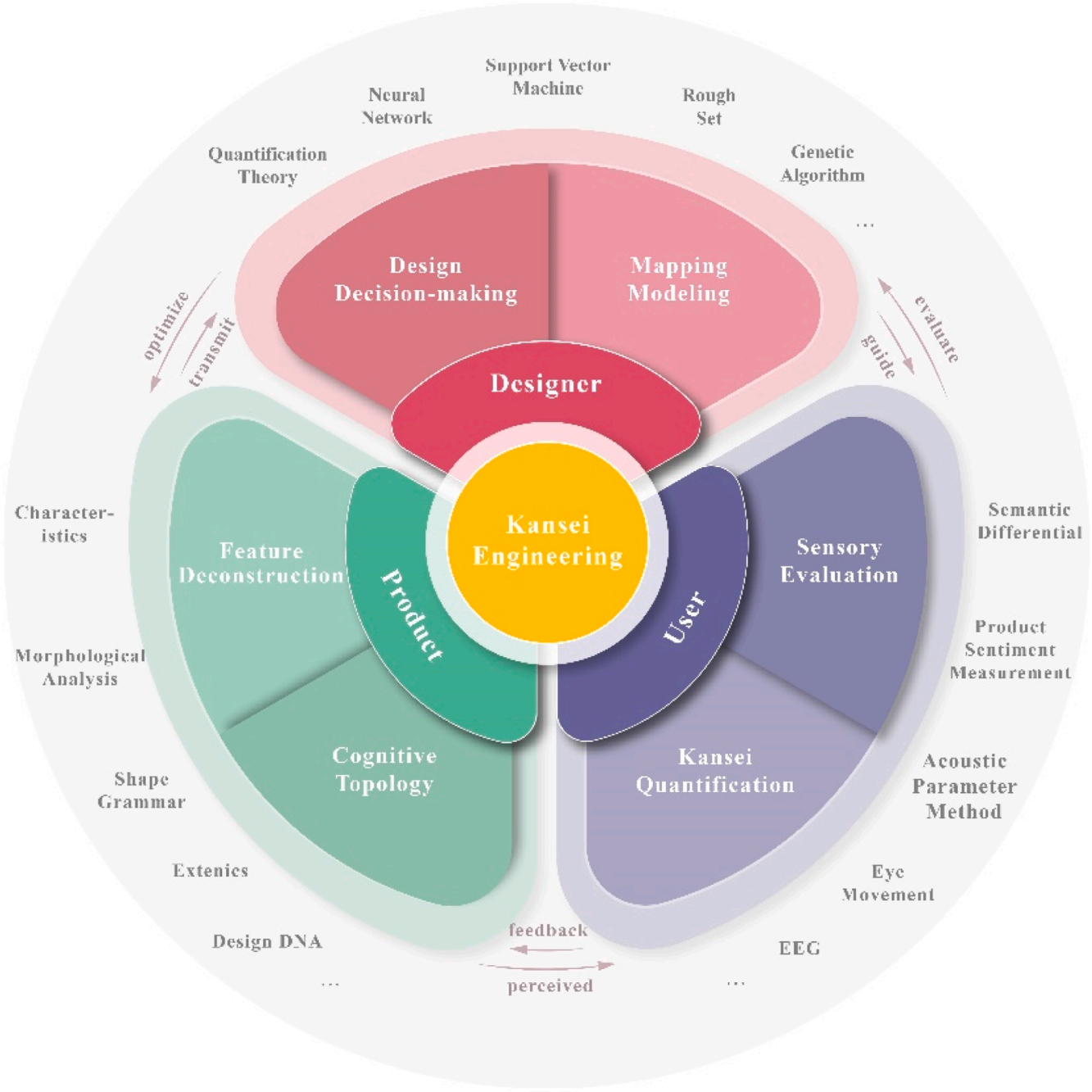


Fig. 9. Scientific challenges involved in Kansei engineering.

① Considering the priority of the complete Kansei, which focuses on the product feature line (usually the boundary line), the general outline of the product appearance is quantitatively described from point to line by defining and adjusting the control points. A typical application is in the creative development of automotive styling, where styling features are defined by the side profile or outline of the front face [68]. ② Considering the Kansei evoked by the overall and detailed design of the product, the appearance of the product is captured using feature composition, feature extraction, and feature analysis. The most common approaches include morphological methods [69], Shape Grammar [70] and extenics [71]. The morphological method decomposes the product form into several basic units, expressed by different characters (including shape, color, and material), and discretely describes the appearance characteristics of the product. Influenced by linguistics, Shape Grammar is an analytical method based on the function of shape. It considers shape as the basic element of products, and the connection between shape elements follows rules and orders. Extenics describes an object according to the primitive principle and considers the combination of the object, object features, and feature levels to create an effective analysis mechanism. For example, a research group defines the design features of a washing machine based on the extension primitive principle and constructed a parameterized product sample space. ③ For product identification, the biological theory of genetic inheritance and variation has been integrated into the design. Based on the characteristics of the product shape, the design DNA is further extracted to characterize the mechanism and rules of shape evolution of the product family [72].

Owing to the infinite variety of products, a universal system theory and tools for deconstructing design features have not yet been developed. Existing methods exhibit typical object dependencies. For example, the ideas for deconstructing the morphological features of a computer mouse and bicycle are completely different. Existing research on KE mainly focuses on consumer goods, and discrete and qualitative methods for deconstructing design features in this product domain are sufficient. However, for complex products in the industrial sector, particularly in the aerospace sector, there remains a lack of systematic parametric methods and tools for deconstructing features in complex dynamic environments.

With the emergence of Artificial Intelligence Generated Content (AIGC) technology, it can quickly generate a large number of rich design feature images through illustrative texts and product images, which not only has higher efficiency and lower output cost, but also has a more comprehensive scope of connotation than the manual method. It provides a new possibility for the understanding, expression and deconstruction of product design features [73].

4.2. Quantitative characterization of Kansei

The Kansei experienced by the user is subtle and complex. It can be perceived but is difficult to express in language [74,75]. The complexity of Kansei necessitates the assessment and evaluation of its quality to be both indirect and partial. Kansei research often achieves the goal of quantification by mastering the elements associated with Kansei [76]. There are four main categories of representative Kansei approaches: behavior, speech, facial expression, and physiological response [77]. Classical quantitative techniques and tools can be divided into physiological and self-reported measurements [78]. Physiological measurements directly measure changes in the human physiological index with special experimental devices such as eye tracking, electroencephalogram, electrodermal activity, electrocardiogram, and electromyography [79]. Self-assessment measurements are based on the respondents' understanding of the predefined rating scale and making corresponding judgments. The most common tools include the semantic differential method based on a linguistic scale [80], product emotion measurement based on animated scale [81], and auditory parameter method based on auditory scale [82].

With the development and application of the new generation of information technologies, big data has undermined the source, composition, and processing methods of data. The quality of Kansei and the requirements of user information can be effectively quantified through the collection, storage, and calculation of big data. For example, a study proposed a design method based on online review data, feature coding of optimization targets by *K*-means, and optimization iterations based on NSGA-II [83]. Based on data farming theory and multi-agent modeling technology, Chen investigated the development of user interest in social networks [84].

Because of the differences in principles, the quantitative tools of Kansei have their own advantages and disadvantages. Self-reporting is suitable for quantifying momentary perceptual feedback and can distinguish the specific composition of subtle Kansei. Physiological measurements and big data intelligence can better track changing trends and record the development process of user and product interaction feedback based on time-series characteristics. However, considering the subtle changes in various physiological indicators induced by these products, physiological measurement methods typically require a comprehensive reference for multiple measurement methods. Different interactive interfaces must be considered for different product objects as well as the fusion of analysis technologies for different sensory channels and heterogeneous data from multiple sources.

4.3. Construction of the Kansei decision system

The Kansei quality obtained by organizing user experiments is obtained from sampling. How to derive global trends and laws from local information and how to deal with the connection between user requirements and product design characteristics to draw reasonable scientific conclusions can be regarded as dealing with regression/function estimation problems. Several tools can be used to build decision models, such as quantitative theory I [85], neural networks [86], support vector machines [87], and rough sets [88]. These tools are used to create a mapping model between the user's emotional feedback and product design characteristics to predict the degree of user preference for product design proposals. The opposite approach involves the use of search algorithms to generate new product design suggestions, for example, the most commonly used genetic algorithm method [89,90].

With the continuous exploration of data application patterns, data intervention in the design industry has gone beyond the scope of statistical prediction and estimation of limited samples. Design decisions based on data intelligence provide a new approach in KE. Tao et al. proposed a digital twin-based method and model for product design to solve the problems of data isolation, scattering, and low efficiency in the manufacturing industry [91]. Kang et al. updated web products according to user behavior data to train machine-learning models [92]. Ma et al. proposed a data-driven predictive model for product-family design (PDPFD), to determine the optimal structure of a product family by capturing customers' uncertain preferences in the market [93].

KE theory is characterized by the openness and involvement of all stakeholders. The possible methods and tools are not limited to those mentioned above and are constantly being supplemented and expanded. The construction of decision models has special algorithm tools in the methodology itself coupled with the existing theories or methods in other disciplines as a reference.

5. Conclusion

This study reviewed the applications of KE as a comprehensive and inclusive research and practice discipline across various vehicle fields. It has been observed that KE offers substantial opportunities and valuable insights for the design and optimization of such products. This study systematically analyzed 562 research articles on the KE in the vehicle field, and the main contributions are summarized as follows:

(1) Using visual statistical analysis, this study systematically summarized and sorted out the development, evolution, and research trends of KE applications in the vehicle field. The research content of the KE in the vehicle field was systematically expounded in three dimensions: land, air, and water vehicles.
(2) Using comprehensive statistical analysis, core journals, core countries, and core authors were identified and visually represented.
(3) This study proposed the core technologies of KE from three perspectives: deconstruction of product design features, quantitative characterization of user Kansei, and construction of the Kansei decision-making system. It systematically elucidated the methods and challenges of Kansei Engineering in scientific research and practical applications, and provides theoretical foundations and technical support for future design and development.
(4) Based on the preceding analysis, the future development trends of the KE in this domain can be categorized into three primary directions:

① Apply the KE to a broader range of transportation product designs and explore its value for transportation infrastructure, components, and related services. This approach can significantly enhance the vitality of the KE. The authors and their research team have already conducted certain studies, such as proposing a multimodal human-machine data-driven service design method centered on the check-in experience.
②Consider the impact of dynamic and complex real-world environments on research outcomes, and propose new measurement paradigms and research methods to increase the practical value of the studies. Although there has been progress in research on automotive product design, gaps remain in the aviation field that need to be addressed.
③ Utilize KE methods to assist and optimize the design process of designers. Existing studies predominantly use KE to evaluate existing products or services, with insufficient attention paid to factors such as human-machine effectiveness, cognitive load, and completion rate during the designers' development process. Enhancing the designer sensibility can improve design efficiency and outcomes.

CRediT authorship contribution statement

Weihua Lu: Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Chenyan Ye:** Writing – original draft, Visualization, Software, Investigation. **Yingying Fang:** Writing – original draft, Visualization, Software, Investigation. **Vanja Čok:** Writing – review & editing. **Jean-François Petiot:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Weihua LU reports financial support was provided by the National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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